

Final JavaScript Project Assignment

Modern Web & App Development

Course: Full Stack Foundation Course

Instructor: Sir Talal Ahmed

Final Project Assignment (Mandatory)

A Message from Your Instructor

Dear Students,

Over the past two months, you have taken your first steps into the world of programming. You started with little or no knowledge of JavaScript, and together we explored the foundations of one of the most powerful programming languages used in modern web development.

Throughout this course, you have learned:

- Variables
- Data Types
- Operators
- Conditional Statements
- Loops
- Functions
- Arrays and Array Methods
- Objects
- DOM Manipulation
- Events
- Local Storage
- JavaScript Logic and Problem Solving

This assignment represents everything you have learned during this course. It is designed to test your understanding, strengthen your problem-solving skills, and prepare you for the next stage of your journey.

Remember, this is **not the end of your learning—it is only the beginning**. Great developers are made through consistent practice, curiosity, and dedication. Continue building projects, solving problems, and writing code every day. The more you practice, the more confident and skilled you will become.

I believe every one of you is capable of completing this project successfully. Give it your best effort, write your own code, and take pride in what you build.

I wish you all the very best.

Project Importance

This is your **Final JavaScript Project**, and it is **mandatory** for every student enrolled in this course.

This project carries significant weight in your overall evaluation and is required for the successful completion of the JavaScript module.

Students who do not submit this project will not be eligible for certification.

No exceptions will be made after the submission deadline. Therefore, complete your project on time and submit it according to the instructions provided.

Please do not contact the instructor later requesting certification if this mandatory project has not been submitted.

Take this assignment seriously and give it your full attention.

Project Objective

Develop a **Library Management System** using **Pure (Vanilla) JavaScript**.

Your application must implement complete **CRUD (Create, Read, Update, Delete)** functionality while using **Local Storage** to permanently store data inside the browser.

The purpose of this assignment is to demonstrate your understanding of JavaScript fundamentals by developing a complete real-world application without using any JavaScript frameworks or libraries.

Project Requirements

Your Library Management System must include the following functionality.

1. Add Book (Create)

The user should be able to add a new book.

Each book must contain the following information:

- Book Title

- Author Name
- Category / Genre
- ISBN Number or Book ID
- Publication Year
- Availability Status (Available / Issued)

Requirements:

- Validate all required fields.
 - Prevent empty submissions.
 - Save the book into Local Storage.
 - Display the newly added book immediately.
-

2. Display Books (Read)

Display all books inside a clean table, cards, or list.

Each record should display:

- Book Title
- Author
- Category
- ISBN / Book ID
- Publication Year
- Availability Status

All saved books must automatically appear after refreshing the browser.

3. Update Book (Update)

Users must be able to edit any existing book.

They should be able to update:

- Book Title
- Author
- Category
- Publication Year
- Availability Status

After updating:

- The UI should refresh immediately.
- Local Storage must also be updated.

4. Delete Book (Delete)

Each book should have its own **Delete** button.

When deleted:

- The record should disappear immediately.
- Local Storage should also update automatically.

5. Local Storage (Mandatory)

Your project **must** use Local Storage.

The application should:

- Save all book records.
- Automatically load records after page refresh.
- Preserve data until it is deleted by the user.

Projects that do not use Local Storage will be considered incomplete.

JavaScript Concepts That Must Be Used

Your project should clearly demonstrate your understanding of:

- Variables
 - Data Types
 - Arrays
 - Objects
 - Functions
 - Loops
 - Conditional Statements
 - Array Methods
 - DOM Manipulation
 - Event Listeners
 - Local Storage
 - String Methods (where applicable)
-

HTML & CSS Guidelines

The primary objective of this assignment is to assess your **JavaScript** skills.

Therefore:

- You may design your own HTML and CSS.
- You may use AI tools to generate or improve your HTML and CSS.

However:

All JavaScript logic must be written by you.

You are **not allowed** to use AI to generate your JavaScript logic.

During project evaluation, you may be asked to explain any part of your code.

If you cannot explain your JavaScript implementation, your marks may be reduced, and the project may be considered copied.

Code Quality Requirements

Your JavaScript code should be:

- Properly indented
- Clean and readable
- Organized into reusable functions
- Well-structured
- Free from unnecessary repetition

Use meaningful variable names.

Good Examples:

- bookTitle
- authorName
- publicationYear
- availabilityStatus

Avoid names such as:

- a
- b
- x

- data1
-

Folder Structure

Your submission should follow the structure below:

```
Project Folder
|
├─ index.html
├─ style.css
├─ app.js
└─ assets/
```

Bonus Features (Optional)

Students who complete all required functionality may implement additional features such as:

- Search books
- Filter by category
- Sort books alphabetically
- Display total number of books
- Confirmation dialog before deleting
- Responsive Design
- Attractive animations
- Dark Mode
- Success/Error notification messages

These features are optional and may earn additional marks.

Submission Requirements

Every student must submit:

- Complete Project Folder
- GitHub Repository Link
- Live Hosted Link (GitHub Pages, Netlify, Vercel, etc.)

Incomplete submissions may not be accepted.

Evaluation Criteria

Criteria	Marks
CRUD Functionality	30
JavaScript Logic	25
Local Storage	15
Code Structure & Readability	10
DOM Manipulation	10
UI Functionality	5
Bonus Features & Creativity	5
Total	100

Important Instructions

- This is an individual assignment.
- Every student must complete the project independently.
- Copying another student's project is strictly prohibited.
- HTML and CSS may be created with AI assistance.
- JavaScript logic must be written entirely by the student.
- You should fully understand every line of code you submit.
- You must be prepared to explain your project during evaluation.
- Failure to explain your JavaScript implementation may result in reduced marks.

Deadline

Submission Deadline: Thursday 7, 2026

Late submissions may not be accepted.

Final Note

This assignment is more than just a project—it is proof of everything you have learned over the past two months.

Take your time, think before you code, solve problems independently, and don't be afraid to make mistakes. Every mistake is an opportunity to learn.

This project is an important milestone in your journey as a programmer. Complete it with honesty, dedication, and confidence.

I am proud of the effort you have shown throughout this course, and I look forward to seeing the projects you create.

I wish you success in this assignment and in your future career as a software developer.

Best wishes,

Sir Talal Ahmed

Instructor – Modern Web & App Development